ELECTRODYNAMICS II · SEMESTER 5

Electromagnetic waves

Module 3 · Prepare

Start with the idea. Learn the mathematics. Work the questions. Check the result.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

How to use this book

P means prerequisite, M means mathematics, T means topic and Q means the numbered practice question. Topic codes restart in each module. Read P1 and M1 first, then follow the topics in order.

REMIND ME

A fact from school that we will use again.

MATHS TOOL

The meaning of a symbol and the operation it asks you to do.

THE IDEA

A physical explanation before the equation.

RESULT

The statement to carry into an exam.

EXAM TRAP

An assumption, sign or unit that changes the answer.

CHECK YOURSELF

Cover the answer and try. Then explain why your answer makes sense.

One rule for this module

> **KEEP THIS IN MIND**
>
> A travelling electromagnetic disturbance carries linked transverse E and B fields, energy and momentum.

The whole module in five sentences

1. A travelling waveform keeps its shape while its argument shifts in space and time.
2. Maxwell's equations make E and B obey wave equations in empty space.
3. The fields are transverse, in phase and linked by the propagation direction.
4. The Poynting vector tells how electromagnetic energy crosses a surface.
5. A linear transparent medium changes wave speed and impedance; waves still obey the same geometric and conservation rules.

Syllabus route

Units 9–14 · 8 hours · 18 theory marks. Griffiths, 5th ed.: §§9.1.1, 9.1.2, 9.1.4, 9.2.1, 9.2.2, 8.1.2, 9.2.3, 9.3.1.

Definitions and derivations are independently explained from the named textbook sections. Worked numerical values are original practice unless a past-paper tag is printed.

Topic route

Unit 9 → T1–T3; unit 10 → T4; unit 11 → T5; unit 12 → T6; unit 13 → T7–T8; unit 14 → T9.

T1 · Travelling waves in one dimension
T2 · Sinusoidal waves and phase
T3 · Polarization of a wave
T4 · Maxwell’s vacuum wave equation
T5 · Plane-wave geometry and linked fields
T6 · Poynting’s theorem
T7 · Wave energy and intensity
T8 · Momentum and radiation pressure
T9 · Waves in linear media

P1 School physics bridge

From a moving ripple to a field wave

A wave can be described by a function f(z,t): the value at position z and time t. A string wave may describe displacement in metres. An electromagnetic wave describes E in V/m and B in tesla. The travelling pattern need not carry material along its direction.

One cycle, three measures

Wavelength λ is the distance between neighbouring crests. Period T is the time between successive crests passing one point. Frequency $f = 1/T$ is measured in hertz. Speed is $v = \lambda f$. Angular frequency $\omega = 2\pi f$ counts radians per second; wave number $k = 2\pi/\lambda$ counts radians per metre.

Energy and momentum

Energy measures the capacity for work (joule). Power is energy per time (watt). Intensity is average power per area (W/m²). Momentum has units kg m/s; changing it requires force. A wave can exert force by transferring momentum even though it carries no material medium through space.

Field laws recalled

In vacuum with no charges/currents: $\nabla \cdot E = 0$, $\nabla \cdot B = 0$, $\nabla \times E = -\partial_t B$ and $\nabla \times B = \mu_0 \epsilon_0 \partial_t E$. The vacuum speed is $c = 1/\sqrt{\mu_0 \epsilon_0} \simeq 3.00 \times 10^8$ m/s.

M1 Mathematics from the beginning

Reading a travelling cosine

Consider $E_x = E_0 \cos(kz - \omega t + \delta)$. E_0 is its peak, not its average. The expression in brackets is the phase. A crest keeps that phase constant. Thus $k\Delta z - \omega \Delta t = 0$, giving speed $\Delta z/\Delta t = \omega/k$ toward +z. A plus sign before ωt gives travel toward −z for positive k and ω.

Differentiation with the chain rule

$d(\cos u)/du = -\sin u$. If $u = kz - \omega t$, then $\partial_z \cos u = -k \sin u$ and $\partial_t \cos u = +\omega \sin u$. Differentiating twice gives $\partial_z^2 \cos u = -k^2 \cos u$ and $\partial_t^2 \cos u = -\omega^2 \cos u$. A partial derivative changes just one variable.

The wave equation is a test

Insert those derivatives into $\partial_z^2 E = (1/v^2)\partial_t^2 E$. Cancel the common cosine and amplitude to obtain $k^2 = \omega^2/v^2$. This verifies the wave if $v = \omega/k$. The second derivative measures curvature: the rule connects spatial bending of the graph with its acceleration in time.

Vector tools

$\hat{x} \times \hat{y} = \hat{z}$; reverse the order and change sign. For transverse E in a wave travelling along $\hat{k}$, magnetic field is $B = (1/v)\hat{k} \times E$. The vector identity $\nabla \times (\nabla \times E) = \nabla(\nabla \cdot E) - \nabla^2 E$ converts two curls to a wave equation; ∇^2 means the sum of second spatial derivatives of each component.

Averages and root mean square

Over one cycle, mean cosine is zero while mean squared cosine is 1/2. This follows from $\cos^2\theta = (1 + \cos 2\theta)/2$: the second term averages to zero. Thus $E_{\text{rms}} = \sqrt{\langle E^2 \rangle} = E_0/\sqrt{2}$. Intensity involves the square, so it remains positive even though the signed field average is zero.

Energy accounting identity

$\nabla \cdot (E \times B) = B \cdot (\nabla \times E) - E \cdot (\nabla \times B)$. This is a vector product rule. It lets Maxwell's curls turn into the divergence of an energy-flow vector. The divergence theorem then converts local accounting to total energy crossing the boundary of a volume.

T1 Travelling waves in one dimension

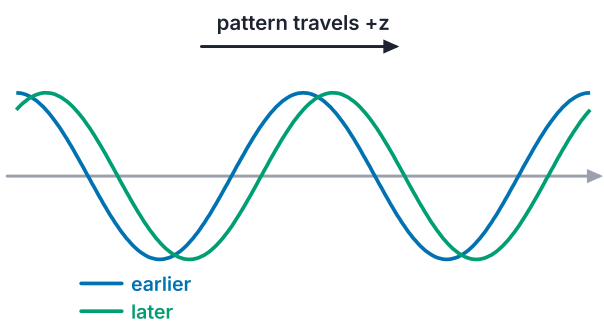
> Purpose: The minus sign in f(z−vt) makes a fixed feature move toward increasing z.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

The minus sign in f(z−vt) makes a fixed feature move toward increasing z.



A cosine profile shifts toward +z as time increases. Field oscillation and pattern travel are different motions.

1. Follow a fixed value of the argument $z - vt$.
   Why: A crest keeps its position in the waveform.

2. If time increases by Δt, z must increase by vΔt.
   Why: This keeps the argument constant.

3. Differentiating any smooth f twice gives the wave equation.
   Why: The chain rule supplies a factor v^2 in the time derivative.

RESULT

$f(z-vt)$ travels toward +z; $f(z+vt)$ toward −z for v>0; $\partial_z^2 f = v^{-2}\partial_t^2 f$.

EXAM TRAP

A general solution can contain waves in both directions; one sign describes one travelling contribution.

CHECK YOURSELF

Which way does $\cos(2z+6t)$ travel?
Answer: Toward −z at 3 m/s if SI units are used.

This topic supports Q1, Q2, Q21.

Work it through · Q1

Skill: Direction from phase. Model · syllabus-based

Determine the direction and speed of $f(z,t) = A\cos(4z+12t)$, with SI coordinates.

Keep $4z+12t$ constant: $4\,dz/dt + 12 = 0$, so $dz/dt = -3$ m/s. A constant-phase point travels toward decreasing z.

CHECK THE ENDPOINT

Speed 3 m/s toward −z.

Work it through · Q2

Skill: Meaning of wave equation. Model · syllabus-based

Write the one-dimensional wave equation and interpret its two derivatives.

$\partial_z^2 f = v^{-2}\partial_t^2 f$. The left side is spatial curvature; the right side is time acceleration divided by speed squared. It constrains how a disturbance propagates, rather than specifying one particular shape.

CHECK THE ENDPOINT

$\partial_t^2 f = v^2 \partial_z^2 f$.

Work it through · Q21

Skill: Arbitrary travelling waveform verification. Model · syllabus-based

Show directly that any twice-differentiable $f(z - vt)$ solves the one-dimensional wave equation.

1. Set $s = z - vt$. Then $\partial_z f = f'(s)$ and $\partial_z^2 f = f''(s)$.
   Why: The derivative of s with respect to z is one.
2. $\partial_t f = -v f'(s)$ and $\partial_t^2 f = v^2 f''(s)$.
   Why: Each time derivative contributes −v.
3. Thus $\partial_t^2 f = v^2 \partial_z^2 f$.
   Why: Both sides equal $v^2 f''(s)$.

4. The shape is unchanged as s remains constant along z=vt+constant.
   Why: This also explains the direction of propagation.

CHECK THE ENDPOINT

Every smooth profile f(z−vt) is a right-travelling solution.

T2 Sinusoidal waves and phase

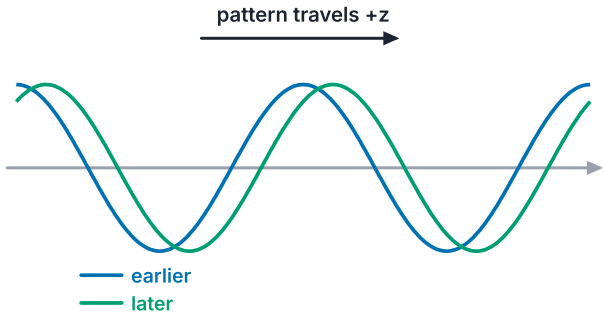
Purpose: A sinusoid repeats in both space and time; k and ω encode those repetitions.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

A sinusoid repeats in both space and time; k and ω encode those repetitions.



A cosine profile shifts toward +z as time increases. Field oscillation and pattern travel are different motions.

1. A spatial shift λ must change phase by 2π.
   Why: That is one complete cycle.
2. A time interval T changes phase by 2π at fixed z.
   Why: This defines period and frequency.
3. A phase offset δ moves the location of crests but does not change speed.
   Why: Speed depends on the ratio ω/k.

RESULT

$E = E_0 \cos(kz - \omega t + \delta)$; $k = 2\pi/\lambda$, $\omega = 2\pi f$, $v = \omega/k$.

EXAM TRAP

Use angular frequency in rad/s with k in rad/m; do not mix ω with ordinary f.

CHECK YOURSELF

If frequency doubles at fixed speed, what happens to λ?

Answer: It halves.

This topic supports Q3, Q4, Q22.

Work it through · Q3

Skill: Frequency wavelength wave number. Model · syllabus-based

A vacuum wave has f=100 MHz. Find λ, k and ω.

$\lambda = c/f = 3.00$ m; $k = 2\pi/\lambda = 2.094$ rad/m; $\omega = 2\pi f = 6.283 \times 10^8$ rad/s. MHz means 10^6 Hz.

CHECK THE ENDPOINT

λ=3.00 m, k=2.094 rad/m, $\omega = 6.283 \times 10^8$ rad/s.

Work it through · Q4

Skill: Phase difference. Model · syllabus-based

Two points along a sinusoidal wave are separated by λ/4. What is their phase difference at the same time?

$\Delta\phi = k\Delta z = (2\pi/\lambda)(\lambda/4) = \pi/2$ radians. They are a quarter-cycle apart; which leads depends on the ordered positions and propagation direction.

CHECK THE ENDPOINT

Phase separation magnitude $\pi/2$.

Work it through · Q22

Skill: Recovering a waveform from specifications. Model · syllabus-based

Write E for a +z travelling, x-polarized vacuum wave with peak 20 V/m and wavelength 0.60 m, whose field is maximum at z=t=0. Find B.

1. $k = 2\pi/0.60 = 10.472$ rad/m and $\omega = ck = 3.142 \times 10^9$ rad/s.
   Why: Vacuum dispersion is $\omega = ck$.
2. Choose cosine with zero phase: $E = 20\cos(kz - \omega t)\hat{x}$.
   Why: Cosine is one at the specified origin.
3. $B = (20/c)\cos(kz - \omega t)\hat{y}$.
   Why: Use the right-handed plane-wave relation.
4. $B_0 = 6.667 \times 10^{-8}$ T.
   Why: Electric and magnetic amplitudes differ by c.

CHECK THE ENDPOINT

$E = 20\cos(10.472z - 3.142 \times 10^9 t)\hat{x}$ V/m; B has the same phase and +y direction.

T3 Polarization of a wave

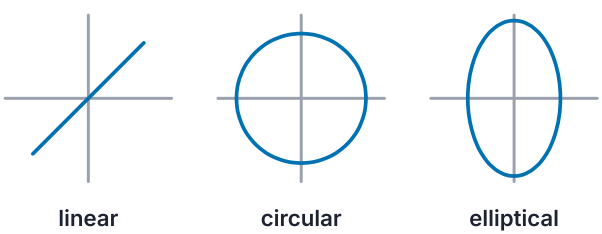
> Purpose: Polarization describes the path traced by the transverse electric-field tip at one location.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

Polarization describes the path traced by the transverse electric-field tip at one location.



Paths traced by the electric-field tip at a fixed position, viewed in the plane perpendicular to propagation.

1. Resolve E along two perpendicular transverse directions.
   Why: Both are allowed by transversality.
2. Equal or opposite phases give a fixed line.
   Why: The component ratio is constant.
3. A quarter-cycle phase difference with equal amplitudes gives a circle.
   Why: Sine and cosine squares sum to one.

4. Unequal amplitudes or a general phase difference give an ellipse.
   Why: The tip traces a closed tilted curve in the transverse plane.

RESULT

Linear: fixed line; circular: equal perpendicular amplitudes in quadrature; elliptical: general perpendicular components.

EXAM TRAP

Handedness labels depend on whether the observer looks with or against propagation; state the viewing convention.

CHECK YOURSELF

Equal amplitudes with zero phase difference: circular or linear?

Answer: Linear, along a 45° line.

This topic supports Q5, Q6, Q7, Q8, Q23, Q24.

Work it through · Q5

Skill: Polarization definition. Model · syllabus-based

Define polarization and explain why a vacuum plane EM wave can be polarized.

Polarization specifies the motion of the E-field tip in the plane transverse to propagation at a fixed point. Two independent transverse components can have different amplitudes and phase relations, giving different polarization states.

CHECK THE ENDPOINT

Polarization concerns transverse E orientation and its time evolution.

Work it through · Q6

Skill: Circular polarization conditions. Model · syllabus-based

What amplitude and phase conditions produce circular polarization from two perpendicular components?

The components have equal nonzero amplitudes and phase difference $\pm\pi/2$ modulo 2π. The magnitude of E is then constant while its tip rotates. Zero phase difference instead gives a straight line.

CHECK THE ENDPOINT

Equal perpendicular amplitudes in quadrature give a circle.

Work it through · Q7

Skill: Linear vector composition. Model · syllabus-based

Classify $E_x = 3\cos\phi$ and $E_y = 4\cos\phi$. Give peak magnitude and orientation.

The component ratio is $E_y/E_x = 4/3$, constant whenever defined. The tip moves on a line at $\tan^{-1}(4/3) = 53.13°$ to +x. Peak magnitude is $\sqrt{3^2 + 4^2} = 5$ in the field units.

CHECK THE ENDPOINT

Linear polarization, peak 5, line angle 53.13°.

Work it through · Q8

Skill: Elliptical special case. Model · syllabus-based

Classify $E_x = 3\cos\phi$, $E_y = 4\sin\phi$ and write the tip's locus.

Eliminate phase using $\cos^2\phi + \sin^2\phi = 1$. This gives $E_x^2/9 + E_y^2/16 = 1$, an ellipse with semi-axes 3 and 4. The fields are in quadrature but amplitudes differ.

CHECK THE ENDPOINT

Elliptical polarization.

Work it through · Q23

Skill: Eliminating phase for polarization ellipse. Model · syllabus-based

For $E_x = a\cos\phi$ and $E_y = b\cos(\phi + \delta)$, derive the polarization ellipse equation.

1. Set $X = E_x/a = \cos\phi$ and $Y = E_y/b = \cos\phi\cos\delta - \sin\phi\sin\delta$.
   Why: Expand the cosine of a sum.
2. Rearrange $Y - X\cos\delta = -\sin\phi\sin\delta$.
   Why: Isolate the sine term.
3. Square and use $\sin^2\phi = 1 - X^2$.
   Why: Eliminate the phase.
4. Collect terms: $X^2 + Y^2 - 2XY\cos\delta = \sin^2\delta$.
   Why: This is the general conic relation.
5. For δ=0 or π it degenerates to a line; for equal a,b and δ=±π/2 it is a circle.
   Why: These are limiting phase and amplitude choices.

CHECK THE ENDPOINT

$E_x^2/a^2 + E_y^2/b^2 - 2E_xE_y\cos\delta/(ab) = \sin^2\delta.$

Work it through · Q24

Skill: Polarization rotation with explicit viewpoint. Model · syllabus-based

At z=0 take $E_x = E_0\cos\omega t$, $E_y = E_0\sin\omega t$. Describe its rotation as viewed from +z looking toward the origin. Avoid an unqualified handedness label.

1. At t=0 the tip is on +x.
   Why: Cosine is one and sine zero.
2. At T/4 it is on +y, at T/2 on −x and at 3T/4 on −y.
   Why: Advance the phase by π/2 each quarter-cycle.
3. Viewed from +z toward the origin, +x to +y is counterclockwise.
   Why: This is the standard xy-plane orientation from its positive normal.
4. The magnitude is $\sqrt{E_x^2 + E_y^2} = E_0$.
   Why: Sine and cosine squares add to one.

CHECK THE ENDPOINT

Circular motion, counterclockwise from the explicitly stated +z viewpoint.

T4 Maxwell’s vacuum wave equation

> Purpose: Two coupled curl equations predict waves travelling at c.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

Two coupled curl equations predict waves travelling at c.

1. Take curl of Faraday's law.
   Why: This introduces curl B after differentiating in time.
2. Use the curl-curl identity and source-free divergence $\nabla \cdot E = 0$.
   Why: The gradient-of-divergence term vanishes.
3. Insert Ampère-Maxwell law.
   Why: It replaces curl B by a time derivative of E.
4. The two minus signs cancel, leaving the wave equation.
   Why: A second time derivative comes from the two coupled laws.

RESULT

$\nabla^2 E = \mu_0\epsilon_0\partial_t^2 E;\ \nabla^2 B = \mu_0\epsilon_0\partial_t^2 B.$

EXAM TRAP

The source-free vacuum derivation does not apply unchanged where charge, current or varying material coefficients are present.

CHECK YOURSELF

What physical speed follows from $\mu_0\epsilon_0$?
Answer: $c = 1/\sqrt{\mu_0\epsilon_0}$.

This topic supports Q9, Q10, Q25, Q26, Q37.

Work it through · Q9

Skill: Vacuum speed prediction. Model · syllabus-based

How does Maxwell theory predict the speed of waves in vacuum?

The source-free curl equations give $\nabla^2 E = \mu_0\epsilon_0\partial_t^2 E$. Comparing with the wave equation gives $c = (\mu_0\epsilon_0)^{-1/2} \simeq 3.00 \times 10^8$ m/s.

CHECK THE ENDPOINT

Vacuum wave speed is determined by μ_o and ε_o.

Work it through · Q10

Skill: Source-free assumption. Model · syllabus-based

State the assumptions needed for the elementary vacuum EM wave equation.

The region is vacuum with no free or bound charges/currents, so ρ=0, J=0, D=ε_oE and B=μ_oH. The vacuum constants are uniform and the fields are differentiable. Sources may exist outside the region.

CHECK THE ENDPOINT

The derivation applies locally in a source-free vacuum region.

Work it through · Q25

Skill: Electric wave equation derivation. Model · syllabus-based

Derive the source-free vacuum wave equation for E from Maxwell's equations.

1. Take curl of $\nabla \times E = -\partial_t B$.
   Why: We want an equation involving E alone.
2. Use $\nabla \times \nabla \times E = \nabla(\nabla \cdot E) - \nabla^2 E = -\nabla^2 E$.
   Why: Source-free electric divergence is zero.
3. The right side becomes $-\partial_t(\nabla \times B) = -\mu_0 \epsilon_0 \partial_t^2 E$.
   Why: Insert Ampère-Maxwell law.
4. Cancel minus signs.
   Why: Both sides carry the same sign.

CHECK THE ENDPOINT

$\nabla^2 E = \mu_0 \epsilon_0 \partial_t^2 E$.

Work it through · Q26

Skill: Magnetic wave equation derivation. Model · syllabus-based

Derive the source-free vacuum wave equation for B, stating the identity and divergence law used.

1. Curl Ampère-Maxwell: $\nabla \times \nabla \times B = \mu_0 \epsilon_0 \partial_t (\nabla \times E)$.
   Why: This eliminates E after substitution.
2. The left is $-\nabla^2 B$ because $\nabla \cdot B = 0$.
   Why: Use the curl-curl identity.
3. Faraday's law makes the right side $-\mu_0 \epsilon_0 \partial_t^2 B$.
   Why: Its minus sign is essential.
4. Cancel signs and identify speed $c = (\mu_0 \epsilon_0)^{-1/2}$.
   Why: Compare with the standard wave equation.

CHECK THE ENDPOINT

$\nabla^2 B = \mu_0 \epsilon_0 \partial_t^2 B$.

Work it through · Q37

Skill: Full vacuum plane-wave construction. Model · syllabus-based

Starting with source-free Maxwell equations, derive the wave speed, transversality and electric/magnetic amplitude relation of a monochromatic plane wave.

1. Curl Faraday and use the identity $\nabla \times \nabla \times E = \nabla(\nabla \cdot E) - \nabla^2 E$.
   Why: This connects curls to spatial curvature.

2. With zero electric divergence and Ampère-Maxwell law, obtain $\nabla^2 E = \mu_0 \epsilon_0 \partial_t^2 E$.
   Why: The fields are source-free vacuum fields.

3. A cosine of phase $k \cdot r - \omega t$ then requires $\omega^2 = c^2 k^2$, with $c = 1/\sqrt{\mu_0 \epsilon_0}$.
   Why: Second derivatives supply $-k^2$ and $-\omega^2$.

4. The divergence laws give $k \cdot E_0 = k \cdot B_0 = 0$.
   Why: The oscillatory fields have no longitudinal component.

5. Faraday gives $k \times E_0 = \omega B_0$, hence $B_0 = (1/c)\hat{k} \times E_0$.
   Why: The curl law fixes orientation, amplitude and phase.

6. For travel +z choose E along x and B along y with the same cosine.
   Why: Then E cross B points along propagation.

CHECK THE ENDPOINT

Vacuum waves travel at c and have transverse in-phase fields with $E_0 = cB_0$.

T5 Plane-wave geometry and linked fields

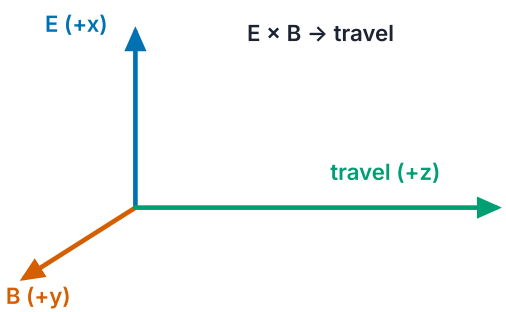
> Purpose: In a source-free plane wave, E, B and propagation are mutually perpendicular.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

In a source-free plane wave, E, B and propagation are mutually perpendicular.



A perspective schematic of three mutually perpendicular directions. E and B oscillate in phase; the propagation vector is not a third oscillating field.

1. For fields depending on phase $\mathbf{k}\cdot\mathbf{r}-\omega t$, the divergence laws require $\mathbf{k}\cdot E=\mathbf{k}\cdot B=0$.
   Why: A nonzero oscillatory longitudinal component would have nonzero divergence.

2. Faraday's law gives $\mathbf{k}\times E=\omega B$.
   Why: Space and time derivatives share the same sinusoidal factor.

3. With $\omega = ck$, B is fixed by E and the travel direction.
   Why: The two fields are not independently chosen.

RESULT

$B_0 = E_0/c$ in vacuum; $\mathbf{B} = (1/c)\hat{\mathbf{k}} \times \mathbf{E}$.

EXAM TRAP

In a travelling vacuum plane wave E and B are in phase, not a quarter-cycle apart.

CHECK YOURSELF

If travel is +z and E points +x at an instant, which way is B?
Answer: +y.

This topic supports Q11, Q12, Q13, Q27, Q28.

Work it through · Q11

Skill: Field handedness. Model · syllabus-based

A wave travels along +x and E points +z at an instant. Which way does B point?

$B = (1/c)\hat{x} \times E$. Since $\hat{x} \times \hat{z} = -\hat{y}$, B points −y.
The check is $E \times B = \hat{z} \times (-\hat{y}) = +\hat{x}$.

CHECK THE ENDPOINT

B points −y.

Work it through · Q12

Skill: Magnetic amplitude. Model · syllabus-based

A vacuum plane wave has electric peak 300 V/m. Find its magnetic peak and relative phase.

$B_0 = E_0/c = 300/(3.00 \times 10^8) = 1.00 \times 10^{-6}$ T. E and B reach their signed extrema together for a travelling plane wave; they are in phase.

CHECK THE ENDPOINT

$B_0 = 1.00$ μT, in phase with E.

Work it through · Q13

Skill: Transversality. Model · syllabus-based

What does transverse mean for a vacuum plane electromagnetic wave?

Both $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to the propagation vector k: $k \cdot E = k \cdot B = 0$. They are also perpendicular to one another. Their oscillations are not along the travel direction.

CHECK THE ENDPOINT

E, B and k form mutually perpendicular directions.

Work it through · Q27

Skill: Deriving the amplitude link. Model · syllabus-based

For $E = E_0 \cos(kz - \omega t)\hat{x}$, use Faraday's law to derive B for a vacuum travelling wave.

1. The y component of curl E is $\partial_z E_x = -kE_0 \sin(kz - \omega t)$.
   Why: Only E_x varies, and only along z.

2. Try $B = B_0 \cos(kz - \omega t)\hat{y}$, so $-\partial_t B_y = -\omega B_0 \sin(kz - \omega t)$.
   Why: The time derivative of cosine contributes +ω before Faraday's minus sign.

3. Equate amplitudes: $kE_0 = \omega B_0$.
   Why: Both sides have the same phase factor.

4. Since $\omega/k = c$, $B_0 = E_0/c$.
   Why: Use vacuum dispersion.

CHECK THE ENDPOINT

B points +y, is in phase, and has peak E_0/c.

Work it through · Q28

Skill: Testing an inconsistent field pair. Model · syllabus-based

Someone proposes a +z vacuum wave with E along +x and B along +y but $B_0 = 2E_0/c$. Can the pair satisfy Maxwell's equations? Explain.

1. Its directions pass the right-hand test, but direction alone is insufficient.
   Why: The fields must satisfy the curl equations.

2. Faraday's law requires $kE_0 = \omega B_0$.
   Why: Derivatives connect the amplitudes.

3. Vacuum wave dispersion requires $\omega = ck$, hence $B_0 = E_0/c$.
   Why: Both conditions must hold together.

4. The proposed factor two violates this unless both amplitudes vanish.
   Why: A nonzero source-free travelling pair has a fixed ratio.

CHECK THE ENDPOINT

No: the proposed amplitude ratio violates Maxwell's curl equations.

T6 Poynting's theorem

> Purpose: Field energy decreases when it flows outward or does work on charges.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

Field energy decreases when it flows outward or does work on charges.

1. Dot Ampère-Maxwell with E and Faraday with B.
   Why: Dot products convert field equations to scalar energy rates.

2. Combine them using the divergence identity for $E \times B$.
   Why: The curls become an outward-flow term.

3. Recognize derivatives of squared fields.
   Why: $E \cdot \partial_t E = \partial_t(E^2)/2$.

4. Interpret $J \cdot E$ as work per volume per time done on charges.
   Why: Electric force transfers field energy to moving charge.

RESULT

$\partial_t u + \nabla \cdot S = -J \cdot E$; $u = \epsilon_0 E^2/2 + B^2/(2\mu_0)$; $S = E \times B/\mu_0$.

EXAM TRAP

Poynting's vector has units W/m²; energy density has units J/m³.

CHECK YOURSELF

If no current is present and field energy decreases inside a volume, where can it go?
Answer: Through the boundary as outward electromagnetic energy flow.

This topic supports Q14, Q15, Q29, Q38.

Work it through · Q14

Skill: Poynting vector units. Model · syllabus-based

Define the vacuum Poynting vector and state its physical meaning and SI unit.

$\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$. It describes electromagnetic power flow per unit area, directed along energy transport. Its units are W/m².

CHECK THE ENDPOINT

S is an energy flux density.

Work it through · Q15

Skill: Work density sign. Model · syllabus-based

Interpret positive $\mathbf{J} \cdot \mathbf{E}$ in Poynting's theorem.

Positive $J \cdot E$ means the field does work on charges at that power per unit volume. It removes electromagnetic field energy locally unless replenished by inflow. In an ohmic resistor it equals σE^2.

CHECK THE ENDPOINT

Positive $J \cdot E$ transfers field energy to matter.

Work it through · Q29

Skill: Energy balance in a region. Model · syllabus-based

A region loses field energy at 5 W while charges inside receive 2 W from the field. Find the net outward electromagnetic power.

1. Integrate Poynting's theorem: $dU/dt + P_{\text{out}} = -P_{\text{matter}}$.
   Why: The divergence theorem converts S to boundary flow.

2. Here $dU/dt = -5$ W and $P_{\text{matter}} = +2$ W.
   Why: Loss of stored energy carries a negative derivative.

3. Thus $-5 + P_{\text{out}} = -2$, giving $P_{\text{out}} = 3$ W.
   Why: Part of the released energy goes to matter and the remainder leaves the region.

CHECK THE ENDPOINT

Net outward power is 3 W.

Work it through · Q38

Skill: Derivation and interpretation of Poynting theorem. Model · syllabus-based

Derive Poynting's theorem in vacuum with charges and currents, then give its integral interpretation.

1. Ampère-Maxwell gives $E \cdot (\nabla \times B) = \mu_0 J \cdot E + \mu_0 \epsilon_0 E \cdot \partial_t E$.
   Why: Take its dot product with E.

2. Faraday gives $B \cdot (\nabla \times E) = -B \cdot \partial_t B$.
   Why: Take its dot product with B.

3. Use $\nabla \cdot (E \times B) = B \cdot \nabla \times E - E \cdot \nabla \times B$.
   Why: This combines the curl terms.

4. Divide by μ_0 and identify $u = \epsilon_0 E^2/2 + B^2/(2\mu_0)$ and $S = E \times B/\mu_0$.
   Why: The squared-field derivatives are twice field dotted with its derivative.

5. Obtain $\partial_t u + \nabla \cdot S = -J \cdot E$.
   Why: The remaining term is work on charges.

6. Integrate over volume: $dU/dt + \oint S \cdot da = -\int J \cdot E \, d\tau$.
   Why: The divergence theorem gives outward surface flow.

7. The field's energy loss equals outward radiation plus work delivered to matter.
   Why: This is local and global energy conservation.

CHECK THE ENDPOINT

Poynting's theorem balances stored energy, boundary energy flux and mechanical/electrical work on charge.

T7 Wave energy and intensity

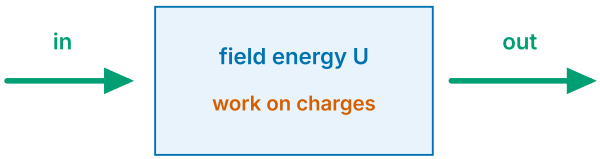
> Purpose: A vacuum travelling wave has equal electric and magnetic energy densities.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

A vacuum travelling wave has equal electric and magnetic energy densities.



Loss = outward flow + work

Poynting accounting: the boundary flux and work on charges account for changes in stored field energy.

1. Use $B = E/c$ in $u_B = B^2/(2\mu_0)$.
   Why: The plane-wave relation fixes the magnetic amplitude.

2. Since $1/(\mu_0 c^2) = \epsilon_0$, $u_B = u_E$.
   Why: The two field energies are equal.

3. Add them and average $\cos^2$ over a cycle.
   Why: Intensity is average, not peak, energy flux.

RESULT

$u = \epsilon_0 E^2$ instantaneously; $S = cu\hat{k}$; $\langle S \rangle = \frac{1}{2}\epsilon_0 c E_0^2 = \epsilon_0 c E_{\text{rms}}^2$.

EXAM TRAP

The factor 1/2 is needed for peak amplitude E_0 but not for rms field.

CHECK YOURSELF

If E_0 triples, what happens to intensity?
Answer: It becomes nine times larger.

This topic supports Q16, Q17, Q30, Q31.

Work it through · Q16

Skill: Equal energy shares. Model · syllabus-based

In a vacuum travelling plane wave, compare electric and magnetic energy densities.

$u_E = \epsilon_0 E^2/2$. Using $B = E/c$ and $1/(\mu_0 c^2) = \epsilon_0$ gives $u_B = B^2/(2\mu_0) = u_E$. Equality holds instantaneously for this travelling plane wave.

CHECK THE ENDPOINT

Each field stores half the total wave energy density.

Work it through · Q17

Skill: Peak and rms distinction. Model · syllabus-based

Express the average intensity of a vacuum sinusoidal wave using peak and rms electric fields.

$I = \frac{1}{2}\epsilon_0 c E_0^2$ and $I = \epsilon_0 c E_{\text{rms}}^2$, with $E_{\text{rms}} = E_0/\sqrt{2}$. The half comes from the cycle average of cosine squared.

CHECK THE ENDPOINT

Use the half only with peak amplitude.

Work it through · Q30

Skill: Intensity from electric peak. Model · syllabus-based

A vacuum plane wave has $E_0 = 100$ V/m. Calculate mean energy density and intensity.

1. $\langle u \rangle = \epsilon_0 E_0^2/2 = (8.854 \times 10^{-12})(10^4)/2 = 4.427 \times 10^{-8}$ J/m³.
   Why: The average squared cosine is 1/2.

2. $I = c\langle u \rangle = 13.28$ W/m².
   Why: A travelling wave carries its energy at c.

3. Equivalently $I = E_0^2/(2\eta_0)$ with $\eta_0 \simeq 376.7$ Ω.
   Why: This provides an independent ratio check.

CHECK THE ENDPOINT

Mean energy density 4.427×10^{-8} J/m³; intensity 13.28 W/m².

Work it through · Q31

Skill: Field amplitude from beam power. Model · syllabus-based

A uniform vacuum beam carries 2 W through area 1.0×10^{-4} m². Find intensity and peak E and B.

1. $I = P/A = 2.0 \times 10^4$ W/m².
   Why: Power is uniformly distributed.

2. $E_0 = \sqrt{2I/(\epsilon_0 c)} = 3.881 \times 10^3$ V/m.
   Why: Invert the peak-field intensity relation.

3. $B_0 = E_0/c = 1.294 \times 10^{-5}$ T.
   Why: The wave is in vacuum.

4. The square root is essential because intensity depends on field squared.
   Why: Doubling power does not double field amplitude.

CHECK THE ENDPOINT

$I=2.0 \times 10^4$ W/m²; $E_0 \approx 3.881$ kV/m; $B_0 \approx 12.94$ μT.

T8 Momentum and radiation pressure

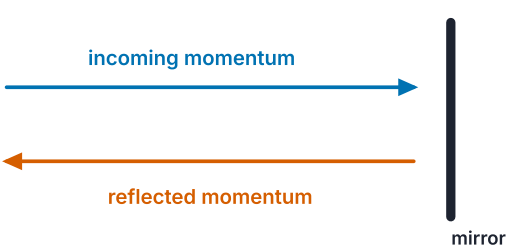
> Purpose: Absorbing or reflecting a wave transfers momentum and therefore exerts force.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

Absorbing or reflecting a wave transfers momentum and therefore exerts force.



Reflection reverses normal wave momentum. The mirror receives twice the incoming momentum per unit time at ideal normal incidence.

1. An absorbed packet of energy U delivers momentum U/c.
   Why: This is the vacuum wave energy-momentum relation.
2. Divide transferred momentum by time and area.
   Why: Force is momentum transfer rate, and pressure is force per area.
3. Perfect reflection reverses the normal momentum.
   Why: The momentum change is doubled.

RESULT

Vacuum: $g = S/c^2$, momentum=energy/c; normal pressure $p = I/c$ for absorption, $p = 2I/c$ for perfect reflection.

EXAM TRAP

The simple I/c and 2I/c formulas assume normal incidence and the stated ideal surface response.

CHECK YOURSELF

A surface is perfectly reflecting rather than absorbing. At equal incident I, pressure changes how?
Answer: It doubles.

This topic supports Q18, Q19, Q32, Q33, Q39.

Work it through · Q18

Skill: Absorption pressure. Model · syllabus-based

What is the pressure on an ideal absorbing surface at normal incidence from a vacuum beam of intensity I?

An absorbed energy U brings momentum U/c. Per time and area, this gives $p = I/c$. The result assumes complete absorption and normal incidence; its units are N/m².

CHECK THE ENDPOINT

$p = I/c$.

Work it through · Q19

Skill: Reflection factor. Model · syllabus-based

Why does an ideal reflector have twice the normal radiation pressure of an ideal absorber at the same incident intensity?

An incident momentum component +U/c becomes −U/c after reflection, so the surface gains 2U/c. Absorption gains only U/c. Dividing by area and time gives $p = 2I/c$.

CHECK THE ENDPOINT

Momentum reversal doubles the transfer.

Work it through · Q32

Skill: Radiation force calculation. Model · syllabus-based

A 3 W beam hits a surface normally in vacuum. Find force for complete absorption and perfect reflection.

1. Incident energy per second is P, so incident momentum per second is P/c.
   Why: A vacuum wave has momentum U/c.
2. Absorption gives $F = P/c = 1.0 \times 10^{-8}$ N.
   Why: All incident momentum is retained.
3. Reflection gives $F = 2P/c = 2.0 \times 10^{-8}$ N.
   Why: Normal momentum reverses.
4. Both forces point into the surface along the incoming beam.
   Why: The surface receives forward momentum.

CHECK THE ENDPOINT

Absorber: 10 nN; reflector: 20 nN.

Work it through · Q33

Skill: Partial reflection pressure. Model · syllabus-based

At normal incidence a surface reflects fraction R of a vacuum beam's intensity and absorbs the remainder, with no transmission. Derive its pressure.

1. The absorbed fraction $(1 - R)I$ transfers pressure $(1 - R)I/c$.
   Why: Absorbed momentum is removed from the beam.

2. The reflected fraction RI transfers $2RI/c$.
   Why: Reflection reverses its normal momentum.
3. Add contributions: $p = (1+R)I/c$.
   Why: Power fractions are mutually exclusive.
4. R=0 and R=1 recover the absorbing and reflecting limits.
   Why: Check the result against known endpoints.

CHECK THE ENDPOINT

$p = (1+R)I/c$ for no transmission at normal incidence.

Work it through · Q39

Skill: Wave energy momentum and pressure synthesis. Model · syllabus-based

Derive the vacuum plane-wave energy/intensity relations and use momentum transport to explain radiation pressure for an absorber and reflector.

1. The plane-wave amplitude relation gives $u_B = u_E = \epsilon_0 E^2/2$.
   Why: Substitute B=E/c into magnetic energy.
2. Total $u = \epsilon_0 E^2$ and $S = cu\hat{k}$.
   Why: Compute the Poynting vector.
3. For sinusoidal peak E_0, $I = \epsilon_0 c E_0^2/2$.
   Why: Average cosine squared.
4. Vacuum momentum density is $g = S/c^2$, so momentum carried by energy U is U/c.
   Why: Divide energy density by c.

5. For normal complete absorption, momentum delivered per time per area gives $p = I/c$.
   Why: Pressure is momentum flux.

6. Perfect reflection reverses the normal momentum and gives $p = 2I/c$.
   Why: The momentum change is twice the incoming value.

7. These formulas refer to vacuum incident waves and ideal stated surface behaviour.
   Why: Inside material, field/matter momentum requires additional care.

CHECK THE ENDPOINT

Energy and momentum transport jointly predict measurable radiation pressure.

T9 Waves in linear media

> Purpose: A homogeneous lossless nondispersive medium changes speed and field ratios.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A wave transports a disturbance; it does not mean the medium travels with the pattern. Frequency counts cycles per second and wavelength measures the length of one cycle.

THE IDEA

A homogeneous lossless nondispersive medium changes speed and field ratios.

1. Use D=εE and B=μH in source-free macroscopic Maxwell equations.
   Why: Constitutive constants describe the medium.
2. Repeat the vacuum curl-curl derivation to obtain $\nabla^2 E = \mu\epsilon\partial_t^2 E$.
   Why: Uniform coefficients can be taken outside derivatives.
3. The speed follows by comparison with the wave equation.
   Why: Its coefficient is $1/v^2$.
4. For a travelling plane wave, intensity is $E_0^2/(2\eta)$ and $S = E \times H$.
   Why: H replaces B/μ_0 in the macroscopic energy flux.

RESULT

$v = 1/\sqrt{\mu\epsilon}$; $n = c/v$; $\eta = \sqrt{\mu/\epsilon}$; $H_0 = E_0/\eta$, $B_0 = E_0/v$.

EXAM TRAP

Conduction, absorption, dispersion or anisotropy require a more general treatment. Vacuum momentum/pressure relations are not automatically transferable inside matter.

CHECK YOURSELF

For $\mu_r = 1$ and $\epsilon_r = 4$, find v and η relative to vacuum.

Answer: $v = c/2$ and $\eta = \eta_0/2$.

This topic supports Q20, Q34, Q35, Q36, Q40.

Work it through · Q20

Skill: Medium speed and index. Model · syllabus-based

A lossless nonmagnetic dielectric has $\epsilon_r = 2.25$. Find refractive index and wave speed.

$n = \sqrt{\mu_r \epsilon_r} = \sqrt{2.25} = 1.5$. Thus $v = c/n = 2.00 \times 10^8$ m/s. This uses a homogeneous nondispersive scalar permittivity and μr=1.

CHECK THE ENDPOINT

n=1.5 and $v = 2.00 \times 10^8$ m/s.

Work it through · Q34

Skill: Impedance and field ratio in medium. Model · syllabus-based

A lossless homogeneous medium has $\epsilon_r = 4$, $\mu_r = 1$ and electric peak 60 V/m. Find speed, impedance, B_o, H_o and intensity.

1. $v = c/2 = 1.50 \times 10^8$ m/s and $\eta = \eta_0/2 \simeq 188.4$ Ω.
   Why: Use the square-root constitutive relations.

2. $B_0 = E_0/v = 4.0 \times 10^{-7}$ T and $H_0 = E_0/\eta = 0.3185$ A/m.
   Why: The ratios are B/E=1/v and E/H=η.

3. $I = E_0^2/(2\eta) = 9.555$ W/m².
   Why: Average the product of in-phase sinusoidal E and H.

CHECK THE ENDPOINT

v=1.50×10^8 m/s, η≈188.4 Ω, B_o=0.40 μT, H_o≈0.3185 A/m, I≈9.555 W/m².

Work it through · Q35

Skill: Frequency across an interface. Model · syllabus-based

A 600 nm vacuum wave enters a stationary transparent nonmagnetic medium with n=1.5. Find f, v and wavelength inside. Explain which quantity stays fixed.

1. Vacuum frequency is $f = c/\lambda_0 = 5.0 \times 10^{14}$ Hz.
   Why: Convert nm to metres.

2. Frequency stays the same across a stationary boundary.
   Why: The boundary fields must match at every instant.

3. $v = c/n = 2.0 \times 10^8$ m/s, so $\lambda = v/f = 400$ nm.
   Why: Use the unchanged f with the reduced speed.

CHECK THE ENDPOINT

f=5.0×10^{14} Hz remains fixed; v=2.0×10^8 m/s; λ=400 nm.

Work it through · Q36

Skill: Energy transport speed in linear medium. Model · syllabus-based

Show that S=vu for a travelling plane wave in a homogeneous linear lossless nondispersive medium.

1. $u = (\epsilon E^2 + \mu H^2)/2$.
   Why: Both field energies are reversible under the stated assumptions.
2. Since $H = E/\eta$ and $\eta^2 = \mu/\epsilon$, $\mu H^2 = \epsilon E^2$, hence $u = \epsilon E^2$.
   Why: The wave's electric and magnetic energy shares are equal.
3. $S = EH = E^2/\eta$.
   Why: The fields are perpendicular and form a travelling wave.
4. $S/u = 1/(\epsilon\eta) = 1/\sqrt{\mu\epsilon} = v$.
   Why: Substitute the impedance definition.

CHECK THE ENDPOINT

$\mathbf{S} = vu\hat{k}$ under the stated nondispersive lossless assumptions.

Work it through · Q40

Skill: Propagation in homogeneous linear matter. Model · syllabus-based

Derive the wave equation and field ratios in a source-free homogeneous linear lossless nondispersive medium, and identify the limits of the model.

1. Use $D = \epsilon E$, $B = \mu H$, with no free current or charge.
   Why: Constitutive laws specify an isotropic linear medium.

2. Then $\nabla \times E = -\mu \partial_t H$ and $\nabla \times H = \epsilon \partial_t E$.
   Why: The constants are uniform and time independent.

3. Curl the first and use $\nabla \cdot E = 0$ to get $\nabla^2 E = \mu\epsilon \partial_t^2 E$.
   Why: No spatial derivatives of ε or μ appear in a homogeneous region.

4. Thus $v = 1/\sqrt{\mu\epsilon}$ and $n = c/v = \sqrt{\mu_r \epsilon_r}$.
   Why: Compare with the wave equation.

5. A transverse plane wave has $B = \hat{k} \times E/v$, $H = \hat{k} \times E/\eta$, with $\eta = \sqrt{\mu/\epsilon}$.
   Why: The curl equations determine the ratios.

6. $S = E \times H$, and mean intensity is $E_0^2/(2\eta)$.
   Why: Average the in-phase sinusoidal product.

7. Absorbing, conducting, dispersive or anisotropic media need extended equations/constitutive treatment.
   Why: The simple real scalar constants then no longer suffice.

CHECK THE ENDPOINT

Speed and impedance are separate material properties: v sets B/E, while η sets E/H.

Formula and condition sheet

T1 · Travelling waves in one dimension

$f(z-vt)$ travels toward +z; $f(z+vt)$ toward −z for v>0; $\partial_z^2 f = v^{-2}\partial_t^2 f$.

T2 · Sinusoidal waves and phase

$E = E_0\cos(kz-\omega t+\delta)$; $k = 2\pi/\lambda$, $\omega = 2\pi f$, $v = \omega/k$.

T3 · Polarization of a wave

Linear: fixed line; circular: equal perpendicular amplitudes in quadrature; elliptical: general perpendicular components.

T4 · Maxwell's vacuum wave equation

$\nabla^2 E = \mu_0\epsilon_0\partial_t^2 E$; $\nabla^2 B = \mu_0\epsilon_0\partial_t^2 B$.

T5 · Plane-wave geometry and linked fields

$B_0 = E_0/c$ in vacuum; $\mathbf{B} = (1/c)\hat{\mathbf{k}}\times\mathbf{E}$.

T6 · Poynting's theorem

$\partial_t u + \nabla\cdot S = -J\cdot E$; $u = \epsilon_0 E^2/2 + B^2/(2\mu_0)$; $S = E\times B/\mu_0$.

T7 · Wave energy and intensity

$u = \epsilon_0 E^2$ instantaneously; $S = cu\hat{k}$; $\langle S\rangle = \frac{1}{2}\epsilon_0 c E_0^2 = \epsilon_0 c E_{\text{rms}}^2$.

T8 · Momentum and radiation pressure

Vacuum: $g = S/c^2$, momentum=energy/c; normal pressure $p = I/c$ for absorption, $p = 2I/c$ for perfect reflection.

T9 · Waves in linear media

$v = 1/\sqrt{\mu\epsilon}$; $n = c/v$; $\eta = \sqrt{\mu/\epsilon}$; $H_0 = E_0/\eta$, $B_0 = E_0/v$.

Past-paper route

No question from the supplied local past papers maps directly to this module. All 40 are model questions, guided by the syllabus; no invented PYQ provenance.